

```

--> a=5;b=13;

--> h=0.1; x=a:h:b; y=sqrt(2*cos(x)-1);

--> inttrap(x,y)
ans =

    2.8580057 + 5.4852776i

--> h=0.1; x=a:h:b; y=sqrt(2*log(x)-1);

--> h=0.1; x=a:h:b; y=sqrt(2-x);

--> inttrap(x,y)
ans =

    0. + 20.857699i

--> integrate('7/x','x',1,5)
ans =

    11.266065

--> integrate('sin(x)/(cos(x)*cos(x)+1)','x',%pi/2,%pi)
ans =

    0.7853982

--> integrate('cos(x)','x',-0.5,0.5)
ans =

    0.9588511

```

```

--> deff('y=G(x)', 'y=7/x'); intg(1,5,G);

--> deff('y=G(x)', 'y=7/x'); intg(1,5,G)
ans =

    11.266065

--> deff('y=G(x)', 'y=sin(x)/(cos(x)*cos(x)+1)'); intg (%pi/2,%pi,G)
Предупреждение : переопределение функции: G . Чтобы эт
ans =

    0.7853982

--> h=5;x=50:5:65;

--> y=log10(x)
y =

    1.69897    1.7403627    1.7781513    1.8129134

--> dy=diff(y)
dy =

    0.0413927    0.0377886    0.0347621

--> dy2=diff(y,2)
dy2 =

   -0.0036041   -0.0030265

--> dy3=diff(y,3)
dy3 =

    0.0005777

```

```

--> Y=(dy(1)-dy2(1)/2+dy3(1)/3)/h
Y =

    0.0086775

--> 1/log(10)/x(1)
ans =

    0.0086859

--> Y=(dy-dy2(1:$-1)/2+dy3(1:$-2)/3)/h
операция +: Предупреждение, суммирование матрицы с пустой матрицей даст в рез
Y =

    []

--> (1/log(10))./x(1:$-1)
ans =

    0.0086859    0.0078963    0.0072382

--> y=sin(x)
y =

   -0.2623749   -0.9997552   -0.3048106    0.8268287

--> dy=diff(y)
dy =

   -0.7373803    0.6949446    1.1316393

--> dy2=diff(y,2)
dy2 =

```

```

1.4323249    0.4366947

--> dy3=diff(y,3)
dy3 =

-0.9956301

--> Y=(dy(1)-dy2(1)/2+dy3(1)/3)/h
Y =

-0.3570839

--> -cos(x)
ans =

-0.964966    -0.0221268    0.952413    0.5624539

--> -cos(x(1))
ans =

-0.9649660

--> function f=my(x), f=(x+2)^3+5*x, endfunction;

```